

MEET-COBRA: Machine-learning-based Energy-Efficient proacTive Coordination of handOver, Beamforming and Resource Allocation

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Abstract—Energy-efficient coordination of handover, beamforming and resource allocation (COBRA) in heterogeneous networks (HetNets) remains underexplored, particularly for high-mobility vehicular scenarios under latency constraints. We formulate a stochastic integer non-linear program (SINLP) for COBRA to minimize long-term system transmit power subject to a queue-length-based latency constraint. However, the NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link channel state information (CSI), makes online global COBRA optimization computationally prohibitive. To address this challenge, we propose the MEET-COBRA framework, which uses machine learning (ML) to predict next-frame beam-pair candidates and channel gains from low-overhead historical CSI measurements, and translates these predictions into per-link resource block (RB) demand estimates for globally optimized proactive handover. Meanwhile, each cell performs prediction-guided beamforming and queue-aware resource allocation to keep the realized link efficiencies consistent with the RB-demand estimates, thereby meeting the latency constraint with low transmit power. Simulation results show that, at a medium-to-high traffic load, MEET-COBRA lowers the average transmit power by about 32% and reduces the latency constraint violation probability from 29.47% to 0.68% compared with the Reactive-COBRA baseline, with only lightweight ML inference overhead.

Index Terms—Heterogeneous networks, proactive handover, beamforming, resource allocation, machine learning, energy efficiency.

I. INTRODUCTION

As modern vehicles evolve into intelligent mobile terminals, their efficient communication with infrastructure, other vehicles, and pedestrians becomes essential, which leads to the emergence of connected autonomous vehicles (CAVs) and Vehicle-to-Everything (V2X) communications [1]. CAV applications demand a mix of high throughput for perception and infotainment, ultra-reliable and low-latency communication (uRLLC) for cooperative safety, and high energy efficiency. Such diverse communication quality of service (QoS) requirements are increasingly considered in 3GPP enhanced V2X service specifications and the broader new radio (NR)-V2X ecosystem [2], [3].

In practice, heterogeneous network (HetNet) deployments integrating macro and micro base stations (BSs) across different frequency bands provide a promising evolutionary path for NR-V2X systems to accommodate these requirements [4]. Typically, macro BSs operate in lower-frequency bands to ensure broad and reliable coverage, whereas micro BSs utilize higher-frequency bands to enhance network capacity. Such a two-tier deployment can jointly improve throughput and coverage for V2X scenarios, while reducing data service latency and energy consumption through spatial reuse and shorter links [5], [6].

However, vehicular HetNets face challenges under high mobility, with densely deployed small cells and links sensitive to blockage and beam misalignment. Fast-moving vehicles trigger frequent handovers (HOs), increasing signaling overhead and degrading link performance [7], [8]. Conventional reactive HO mechanisms based on thresholds and time-to-trigger further incur preparation and interruption latency and increase the risks of HO failures and ping-pong events [9]. Meanwhile, beam training and tracking become more error-prone under rapid topology changes and intermittent blockage, leading to excessive alignment overhead and reliability loss, especially near cell edges [10], [11]. Although 5G-Advanced has introduced L1/L2-triggered mobility and fast conditional handover, energy-efficient Coordination of handOver, Beamforming and Resource Allocation (COBRA) in vehicular HetNets is still underexplored [12]–[14].

Recent studies mainly advance three directions. First, proactive handover (PHO) prepares HO in advance by forecasting traffic flows, vehicle trajectories, or channel states, thereby reducing failures and ping-pong events [9], [15]. With machine learning (ML), historical channel state information (CSI) can be leveraged to predict future positions [16], channel conditions [17], and target BSs [18], which further enhances PHO. Second, ML-assisted beamforming (BF) exploits side information such as out-of-band CSI [11] and context information [19] to infer blockage and narrow beam search space while maintaining robustness and throughput [20]. Reinforcement learning can also adapt beam training and tracking policies [21]. Third,

queue-aware resource allocation (RA) prioritizes users by urgency or deadlines and opportunistically harnesses favorable channels to improve throughput and robustness under power budgets [22], [23].

Despite this progress, few studies coordinate PHO with BF and RA in HetNets. Fundamentally, PHO, BF and RA are coupled through the realized link service capability. A PHO decision changes not only the serving BS but also the selected beam-pair and the resulting channel gain, which determine the realized link efficiency and the required radio resources, such as resource blocks (RBs). Without COBRA, PHO may rely on inaccurate link-quality estimates, leading inefficient resource usage, higher transmit power, and degraded latency reliability. Therefore, this paper aims to design a joint optimization framework for PHO, BF and RA in HetNets.

We consider a two-tier vehicular HetNet, and formulate a COBRA optimization problem to minimize the long-term system transmit power subject to a queue-length-based latency constraint. This problem is a two-timescale stochastic integer non-linear program (SINLP): the frame-level PHO decisions are tightly coupled with slot-level BF and RA; the channel states and traffic arrivals are time-varying and only causally observable; and the decision space grows exponentially with the numbers of BSs, users, RBs, and BF codebook sizes. The NP-hardness of the resulting SINLP, together with the high overhead of acquiring timely and accurate per-link CSI, makes online global COBRA optimization computationally prohibitive.

To address these challenges, we develop MEET-COBRA, a prediction-driven coordination framework that jointly optimizes PHO, BF, and RA via a two-timescale closed loop. The key idea is to leverage ML predictions of next-frame beam-pair candidates and channel gains for each link from low-overhead historical CSI measurements, translate these predictions into per-link RB-demand estimates for globally optimized frame-level PHO, and then perform slot-level prediction-guided BF and queue-aware RA at each BS to keep the realized link efficiencies consistent with the RB-demand estimates, thereby meeting the latency constraint with low transmit power.

To evaluate the proposed MEET-COBRA framework, we build a simulation platform that combines realistic vehicular mobility traces generated by SUMO [24] with ray-tracing-based channel data produced by Sionna RT [25]. Simulation results show that MEET-COBRA effectively reduces system transmit power while improving the adherence to latency constraints across different traffic loads.

The rest of the paper is organized as follows: Section II details the system model; Section III formulates the optimization problem; Section IV presents the MEET-COBRA framework; Section V reports simulation results; and Section VI concludes the paper.

II. SYSTEM MODEL

We consider a vehicular communication scenario covered by a two-tier HetNet adopting a control–data-plane separation

architecture. The network consists of one macro BS and M micro BSs. The macro BS serves as the control-plane anchor and makes frame-level PHO decisions, while the micro BSs execute the corresponding data-plane re-association and perform slot-level BF and RA. The BS set is denoted by $\mathcal{M} = \{0, 1, \dots, M\}$, where index $m = 0$ represents the macro BS and $m > 0$ represents a micro BS. BS- m is allocated K_m orthogonal RBs in the frequency band f_m , where each RB has a bandwidth of B_m and a per-RB transmit power of p_m . For simplicity, all micro BSs are assumed to have identical frequency configurations, i.e., $K_m = K_\mu$, $f_m = f_\mu$ and $B_m = B_\mu$ for all $m > 0$, while the macro BS has distinct parameters K_0, f_0, B_0 .

We model the system in discrete time, indexed by slots $n \in \{0, 1, \dots\}$ of duration Δt . At any time slot n , the set of vehicles is $\mathcal{V}(n)$, with cardinality $V(n) = |\mathcal{V}(n)|$. The data-plane connection between BS- m and vehicle- $v \in \mathcal{V}(n)$ is represented by a binary indicator $u_{m,v}(n)$, where $u_{m,v}(n) = 1$ signifies an active data-plane connection and $u_{m,v}(n) = 0$ otherwise. Each vehicle can be associated with at most one BS on the data plane at a time, which imposes the constraint:

$$\sum_{m=0}^M u_{m,v}(n) \leq 1, \quad \forall v \in \mathcal{V}(n). \quad (1)$$

Note that the control-plane connection to the macro BS is always maintained and thus is not explicitly modeled.

The data-plane connection state $u_{m,v}(n)$ is managed by a central PHO server deployed at the macro BS and updated periodically over a time window of N_{HO} slots, which define a PHO frame. The handover configuration for the x -th frame is determined during the preceding $(x-1)$ -th frame and remains static throughout frame x . Therefore, we introduce a frame-based index. Let $\{\cdot\}^{[x]} \triangleq \{\cdot\}(xN_{\text{HO}})$ denote a variable at the start of the x -th frame, and let $\{\cdot\}^{[x]}(i) \triangleq \{\cdot\}(xN_{\text{HO}} + i)$ be the variable at the i -th slot within that frame, where $i \in \mathcal{N}_{\text{HO}} \triangleq \{0, 1, \dots, N_{\text{HO}} - 1\}$. Consequently, the connection state is constant within any frame, which is expressed as:

$$u_{m,v}^{[x]}(i) = u_{m,v}^{[x]}, \quad \forall i \in \mathcal{N}_{\text{HO}}. \quad (2)$$

We further assume the set of vehicles is static throughout each frame, denoted as $\mathcal{V}^{[x]}$.

A. Channel Model

We specify the channel models used to calculate the instantaneous desired-link gains $g_{m,v}^{[x]}(i)$ for both macro-to-vehicle ($m = 0$) and micro-to-vehicle ($m > 0$) links, as well as the interference-link gains when applicable.

1) *Channel model for macro BS*: The macro BS operates in a lower-frequency band (e.g., the microwave band) to provide wide-area coverage. The frame-averaged channel gain between the macro BS and vehicle- v during the x -th frame, denoted as $G_{0,v}^{[x]}$, is modeled as

$$G_{0,v}^{[x]} = \alpha \left(d_{0,v}^{[x]} \right)^{-\beta} 10^{S_v^{[x]}/10}, \quad (3)$$

where α is a power gain factor, β is the path loss exponent, and $d_{m,v}^{[x]}$ is the distance between BS- m and vehicle- v in the x -th frame. The term $S_v^{[x]}$ captures the shadow fading effect, which is modeled as an independent and identically distributed (i.i.d.) zero-mean Gaussian random variable with variance σ_S^2 .

Based on the frame-averaged channel gain $G_{0,v}^{[x]}$, the instantaneous channel gain $g_{0,v}^{[x]}(i)$ in the i -th time slot of the x -th frame is modeled by Rician fading as

$$g_{0,v}^{[x]}(i) = G_{0,v}^{[x]} \left\| \sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{0,v}^{[x]}(i) \right\|^2, \quad (4)$$

where K_R is the Rician K-factor and $h_{m,v}^{[x]}(i) \sim \mathcal{CN}(0, 1)$ is an i.i.d. standard complex Gaussian random variable.

2) *Channel model for micro BS*: Each micro BS operates in a higher-frequency band (e.g., the mmWave band) and is equipped with a uniform linear array (ULA) of M_T antennas, while each vehicle has M_R antennas at this band. For the micro-to-vehicle links, we adopt a geometric stochastic MIMO channel model [26]. The channel between micro BS- m and vehicle- v during the x -th frame is represented by the frame-averaged channel matrix $\mathbf{H}_{m,v}^{[x]} \in \mathbb{C}^{M_R \times M_T}$, which is modeled as a sum of L distinct propagation paths:

$$\mathbf{H}_{m,v}^{[x]} = \sqrt{M_T M_R} \sum_{l=1}^L \gamma_l \mathbf{a}_R(\theta_l) \mathbf{a}_T(\psi_l)^\dagger, \quad (5)$$

where γ_l , θ_l , and ψ_l denote the complex impulse response, angle of arrival (AoA), and angle of departure (AoD) of the l -th path, respectively. The vectors $\mathbf{a}_R(\cdot) \in \mathbb{C}^{M_R}$ and $\mathbf{a}_T(\cdot) \in \mathbb{C}^{M_T}$ are the receive and transmit array response vectors. For the ULAs, the array response vector is given by

$$[\mathbf{a}(\phi)]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi d}{\lambda} i \sin \phi}, \quad i = 0, \dots, N-1, \quad (6)$$

where N is the number of antennas (M_T or M_R), d is the inter-element spacing (typically $d = \lambda/2$), λ is the carrier wavelength, and ϕ is the AoA or AoD (θ_l or ψ_l). The instantaneous channel matrix in the i -th slot of frame- x is modeled by applying small-scale Rician fading as

$$\mathbf{H}_{m,v}^{[x]}(i) = \left(\sqrt{\frac{K_R}{K_R + 1}} + \sqrt{\frac{1}{K_R + 1}} h_{m,v}^{[x]}(i) \right) \cdot \mathbf{H}_{m,v}^{[x]}. \quad (7)$$

Beamforming is employed for the MIMO links by selecting precoding and combining vectors from predefined codebooks. Specifically, we use discrete Fourier transform (DFT) codebooks. A DFT codebook of size N consists of N orthogonal vectors, $\mathcal{C} = \{\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{N-1}\}$, where each codeword $\mathbf{c}_n \in \mathbb{C}^{N \times 1}$ corresponds to a spatial direction. The i -th element of the n -th codeword is

$$[\mathbf{c}_n]_i = \frac{1}{\sqrt{N}} e^{-j \frac{2\pi n i}{N}}, \quad \forall n, i \in \{0, 1, \dots, N-1\}. \quad (8)$$

The transmit codebook $\mathcal{C}_T$ is a DFT codebook of size M_T , and the receive codebook $\mathcal{C}_R$ is a DFT codebook of size M_R .

When micro BS- m serves vehicle- v in the x -th frame, it selects a transmit precoding vector $\mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T$, and vehicle- v selects a receive combining vector $\mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R$. The resulting instantaneous desired-link channel gain is

$$g_{m,v}^{[x]}(i) = \left| \left(\mathbf{w}_{R,m,v}^{[x]}(i) \right)^\dagger \mathbf{H}_{m,v}^{[x]}(i) \mathbf{w}_{T,m,v}^{[x]}(i) \right|^2. \quad (9)$$

For an interfering micro BS- m that does not serve vehicle- v , we do not explicitly model its beam pattern toward vehicle- v . Instead, the corresponding interference-link channel gain is approximated by the Frobenius norm-based average gain of the MIMO channel matrix:

$$g_{I,m,v}^{[x]}(i) \triangleq \frac{1}{M_T M_R} \left\| \mathbf{H}_{m,v}^{[x]}(i) \right\|_F^2, \quad (10)$$

which provides a simple approximation of the effective interference power under random transmit and receive beam directions.

B. Resource Allocation Model

Let $k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}$ denote the number of RBs allocated by BS- m to vehicle- v in the i -th time slot of the x -th frame. Since RBs can only be allocated to connected users, this variable must satisfy

$$k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} k_{m,v}^{[x]}(i), \quad \forall i \in \mathcal{N}_{HO}. \quad (11)$$

Each BS employs orthogonal frequency-division multiple access (OFDMA). Accordingly, the number of RBs utilized by BS- m , $k_m^{[x]}(i)$, is constrained by its total RB capacity K_m :

$$k_m^{[x]}(i) \triangleq \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m. \quad (12)$$

The instantaneous transmit power of BS- m is then given by

$$P_m^{[x]}(i) = p_m k_m^{[x]}(i). \quad (13)$$

The time-averaged system transmit power is the sum of the average power of all BSs:

$$\begin{aligned} \bar{P} &= \sum_{m=0}^M \bar{P}_m = \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{x=0}^{X-1} \sum_{i \in \mathcal{N}_{HO}} \sum_{m=0}^M P_m^{[x]}(i) \\ &= \lim_{X \rightarrow \infty} \frac{1}{X N_{HO}} \sum_{x=0}^{X-1} \sum_{i \in \mathcal{N}_{HO}} \sum_{v \in \mathcal{V}^{[x]}} \sum_{m=0}^M p_m k_{m,v}^{[x]}(i). \end{aligned} \quad (14)$$

Given allocated RBs, the instantaneous signal to interference plus noise ratio (SINR) of the desired-link from BS- m to vehicle- v is given by

$$\text{SINR}_{m,v}^{[x]}(i) = \frac{p_{m,v}^{[x]}(i) g_{m,v}^{[x]}(i)}{F_m N_0 W_{m,v}^{[x]}(i) + I_{m,v}^{[x]}(i)}, \quad (15)$$

where $p_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) p_m$ is the power allocated to the vehicle, $W_{m,v}^{[x]}(i) = k_{m,v}^{[x]}(i) B_m$ is the allocated bandwidth, N_0 is the noise power spectral density, F_m is the noise figure, and $I_{m,v}^{[x]}(i)$ is the inter-cell interference. For the macro BS, $I_{0,v}^{[x]}(i) = 0$ since its frequency band is orthogonal to those of micro BSs. For each micro BS, since the OFDMA-based RB

allocation across different BSs is assumed to be independent and uniformly random, the expected number of overlapping RBs between the set of RBs allocated by BS- m to vehicle- v and the set of RBs utilized by any interfering BS- m' is $\frac{k_{m,v}^{[x]}(i)k_{m',v}^{[x]}(i)}{K_\mu}$. Accordingly, the interference can be modeled as

$$I_{m,v}^{[x]}(i) = \begin{cases} \sum_{m' \neq 0, m} g_{1,m',v}^{[x]}(i) \frac{k_{m,v}^{[x]}(i)k_{m',v}^{[x]}(i)}{K_\mu} p_{m'}, & m > 0, \\ 0, & m = 0. \end{cases} \quad (16)$$

By substituting the expressions for power, bandwidth and interference, the SINR term simplifies to

$$\text{SINR}_{m,v}^{[x]}(i) = \begin{cases} \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{m' \neq 0, m} \frac{k_{m',v}^{[x]}(i)}{K_\mu} p_{m'} g_{1,m',v}^{[x]}(i)}, & m > 0, \\ \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m}, & m = 0. \end{cases} \quad (17)$$

Based on the Shannon-Hartley theorem, the instantaneous achievable rate of this link is expressed as

$$R_{m,v}^{[x]}(i) = W_{m,v}^{[x]}(i) \log_2 \left(1 + \text{SINR}_{m,v}^{[x]}(i) \right). \quad (18)$$

Considering the beam-searching overhead, the instantaneous receive rate of vehicle- v is given by

$$r_v^{[x]}(i) = \sum_{m=0}^M (1 - \eta_{m,v}^{[x]}(i)) R_{m,v}^{[x]}(i), \quad (19)$$

where $\eta_{m,v}^{[x]}(i) \in [0, 1)$ represents the throughput loss factor resulting from beam alignment. Specifically,

$$\eta_{m,v}^{[x]}(i) = \eta N_{m,v}^{[x]}(i), \quad (20)$$

with $N_{m,v}^{[x]}(i)$ denoting the number of beam-search attempts conducted by BS- m and vehicle- v in the i -th slot of frame- x , and η representing the ratio of OFDM symbols consumed by once beam search to the total number of OFDM symbols per slot. Besides, the frame-averaged receive rate of vehicle- v is

$$\bar{r}_v^{[x]} \triangleq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} r_v^{[x]}(i), \quad (21)$$

C. Traffic Model

We model the data traffic for each vehicle- v with a backlog queue, where $q_v(n)$ denotes its length at the beginning of time slot n . The queue length evolves as

$$q_v(n+1) = \max \{q_v(n) - r_v(n)\Delta t, 0\} + a_v(n), \quad (22)$$

where $a_v(n)$ is the amount of data arriving for vehicle- v during time slot- n . This model assumes that the existing backlog $q_v(n)$ is served first at a rate of $r_v(n)$, and new arrivals $a_v(n)$ are enqueued for service in subsequent slots. We denote the average data arrival rate for vehicle- v as λ_v , thus we have $\mathbb{E}[a_v(n)] = \lambda_v \Delta t$.

According to *Little's law* [27], the queuing latency $\tau_v(n) \approx \lambda_v^{-1} q_v(n)$. To keep $\tau_v(n)$ no more than a latency threshold τ_{th} , the queue length should satisfy the following latency constraint:

$$q_v(n) \leq \lambda_v \tau^{\text{th}}. \quad (23)$$

To evaluate the system performance with respect to this latency constraint, we define the latency constraint violation probability U following the approach in [28] as:

$$U = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \frac{\sum_{v \in \mathcal{V}(n)} U_v(n)}{V(n)}, \quad (24)$$

where $U_v(n) = 1$ if $q_v(n) > \lambda_v \tau^{\text{th}}$, and $U_v(n) = 0$ otherwise. This metric serves as a measure of the system latency reliability.

III. PROBLEM FORMULATION

Our goal is to minimize the long-term system transmit power while satisfying the latency-constrained QoS requirements for all vehicles. This joint optimization of HO, BF and RA over different time scales is formulated as the following problem

$$\begin{aligned} \mathbf{P1}: \quad & \min_{\mathbf{u}, \mathbf{k}, \mathbf{W}} \bar{P} \\ \text{s.t.} \quad & \text{C1: } q_v^{[x]}(i) \leq \lambda_v \tau^{\text{th}}, \\ & \text{C2: } \sum_{m=0}^M u_{m,v}^{[x]} \leq 1, \\ & \text{C3: } \sum_{v \in \mathcal{V}^{[x]}} k_{m,v}^{[x]}(i) \leq K_m, \\ & \text{C4: } k_{m,v}^{[x]}(i) = u_{m,v}^{[x]} k_{m,v}^{[x]}(i), \\ & \text{C5: } u_{m,v}^{[x]} \in \{0, 1\}, \\ & \text{C6: } k_{m,v}^{[x]}(i) \in \{0, 1, \dots, K_m\}, \\ & \text{C7: } \mathbf{w}_{T,m,v}^{[x]}(i) \in \mathcal{C}_T, \mathbf{w}_{R,m,v}^{[x]}(i) \in \mathcal{C}_R. \end{aligned} \quad (25)$$

In **P1**, the decision variables are the sets of data-plane connection states $\mathbf{u} \triangleq \{u_{m,v}^{[x]}\}$, RB allocation numbers $\mathbf{k} \triangleq \{k_{m,v}^{[x]}(i)\}$, and BF vectors $\mathbf{W} \triangleq \{[\mathbf{w}_{T,m,v}^{[x]}(i), \mathbf{w}_{R,m,v}^{[x]}(i)]\}$. In our system, $\mathbf{u}$ is decided per frame, while $\mathbf{k}$ and $\mathbf{W}$ are decided per time slot. They are subject to several constraints. C1 is the queue-length-based latency constraint equivalent to Eq.(23). The HO constraints C2 and C5 enforce that each vehicle connects to at most one BS in the data-plane. The RA constraints C3, C4, and C6 limit the total RBs per BS, ensure RBs are only assigned to connected vehicles, and define the integer nature of $k_{m,v}^{[x]}(i)$. Finally, C7 restricts the BF vectors to be chosen from the predefined finite codebooks, which makes the BF decision a combinatorial integer optimization component.

P1 is a two-timescale SINLP. Obtaining its optimal *online* solution is computationally prohibitive due to the following challenges:

1. **Stochastic and causally observed dynamics:** Future system states, including the vehicle set $\mathcal{V}^{[x]}$, the channel matrices $\mathbf{H}_{m,v}^{[x]}(i)$, and the traffic arrivals $a_v(n)$, are time-varying and only causally observable. Moreover, current

decisions have lasting impacts on future power consumption and latency performance. Therefore, the long-horizon online optimization of **P1** is fundamentally challenging.

2. **Combinatorial decision space and NP-hardness:** All decision variables in **P1** are discrete and jointly define a high-dimensional decision space over BS-vehicle association, beam-pair selection, and RB allocation, further complicated by inter-cell interference coupling through the SINR expressions. Consequently, **P1** is an NP-hard integer nonconvex optimization problem whose feasible set grows exponentially with the numbers of BSs, vehicles, RBs, and BF codebook sizes, rendering exhaustive search or exact global optimization impractical.
3. **Coupled two-timescale optimization:** Frame-level HO decisions determine the association topology and constrain the slot-level BF and RA decisions within that frame, while the chosen BF vectors directly affect the desired-link gains, thereby influencing RB allocation and queue evolution. Such cross-timescale dependencies imply that optimizing PHO, BF, and RA in isolation can lead to persistently suboptimal performance, and call for low-complexity joint coordination mechanisms.
4. **Prohibitive CSI measurement overhead:** An online global optimizer would require timely and accurate per-link CSI for all BS-vehicle links and beam-pair options to evaluate channel gains and interference. In high-mobility vehicular scenarios, acquiring such per-link CSI via frequent channel measurements, beam training, and feedback incurs excessive overhead and latency, thus impractical for realization.

In summary, the NP-hardness of **P1**, together with stochastic dynamics, two-timescale coupling, and the prohibitive overhead of real-time per-link CSI acquisition, makes the online global optimization for **P1** computationally prohibitive.

IV. ALGORITHM DESIGN

To solve **P1** with an efficient and computationally feasible online approach, we propose a novel algorithm framework named Machine-learning-based Energy-Efficient proactive COBRA (MEET-COBRA). It leverages ML-predicted beam-pair candidates and channel gains to enable proactive, two-timescale COBRA optimization: the central PHO server determines the next-frame PHO configuration globally, while each BS performs slot-level BF and RA locally to serve its associated vehicles. The following subsections introduce the ML-based prediction module, the *predictive, frame-level* PHO algorithm, the *real-time, slot-level* BF and RA algorithms, and the overall MEET-COBRA procedure.

A. ML-Based Prediction

A neural network (NN) model is developed to predict, for each vehicle, the next-frame optimal beam-pairs, desired-link gains, and interference-link gains to all micro BSs. To reduce CSI feedback overhead and avoid centralized NN inference latency that grows with the number of vehicles, each vehicle

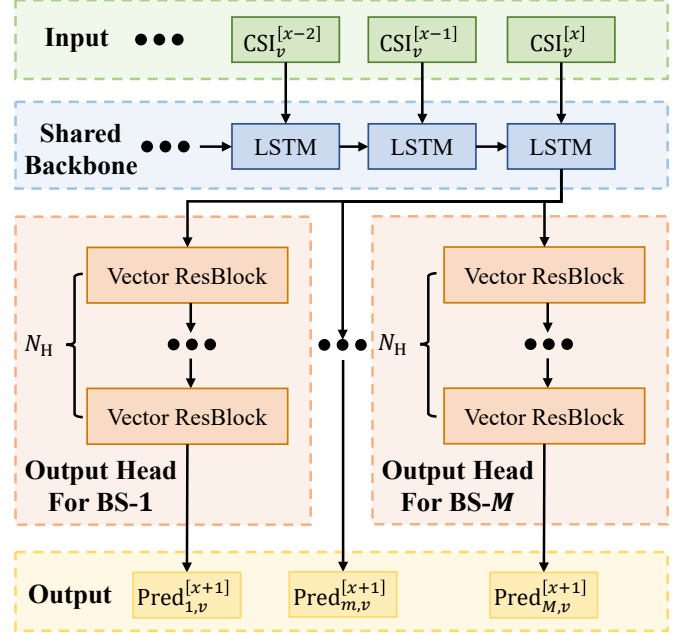


Fig. 1. The architecture of the proposed neural network model for the optimal beam-pair and channel-gain prediction.

performs NN inference locally based on its historical CSI measurements obtained from a low-overhead pilot probing procedure, and reports only the prediction results to the macro BS. The site-specific NN model is shared among vehicles by the macro BS over the control channel. As shown in Fig. 1, the NN consists of an input preprocessing layer, a shared feature-extraction backbone, and M micro-BS-specific output heads.

1) *Input Layer:* The NN input is a sequence of historical CSI measurements of vehicle- v . In each frame, the CSI is measured by a low-overhead pilot probing mechanism based on *pilot superposition*. Specifically, at the beginning of frame- x , all micro BSs simultaneously transmit a series of N_P pilot beams uniformly selected from the transmit codebook $\mathcal{C}_T$. In the p -th pilot probing ($p = 1, \dots, N_P$), the received signal at vehicle- v is the superposition of pilots from all micro BSs, i.e.,

$$\mathbf{y}_v^{[x]}(p) = \sum_{m=1}^M \mathbf{H}_{m,v}^{[x]} \mathbf{w}_{T,m}(p) s(p) + \mathbf{n}_v^{[x]}(p), \quad (26)$$

where $\mathbf{w}_{T,m}(p) \in \mathcal{C}_T$ is the transmit beam used by micro BS- m in the p -th probing, $s(p)$ is the pilot symbol, and $\mathbf{n}_v^{[x]}(p)$ is the received noise. Compared with exhaustive per-BS per-beam pilot sweeping with overhead $O(MM_T)$, pilot superposition reduces the measurement overhead to $O(N_P)$. Accordingly, the CSI measured by vehicle- v in frame- x is defined as

$$\text{CSI}_v^{[x]} \triangleq [\mathbf{y}_v^{[x]}(1)^T, \mathbf{y}_v^{[x]}(2)^T, \dots, \mathbf{y}_v^{[x]}(N_P)^T]^T \in \mathbb{C}^{M_R N_P}, \quad (27)$$

where $\mathbf{y}_v^{[x]}(p) \in \mathbb{C}^{M_R}$ is given in (26). Before being fed into the NN, $\text{CSI}_v^{[x]}$ is converted to a real-valued feature vector by normalizing amplitudes (in dB) and phases, and concatenating

them as $\mathbf{z}_{\text{in},v}^{[x]} \in \mathbb{R}^{2M_R N_P}$. This preprocessing operation is denoted as f_{prep} :

$$\mathbf{z}_{\text{in},v}^{[x]} = f_{\text{prep}} \left(\text{CSI}_v^{[x]} \right). \quad (28)$$

2) *Shared Backbone*: The shared backbone processes the historical CSI sequence of each vehicle to extract a compact representation of its mobility pattern and surrounding radio environment. It consists of a Long Short-Term Memory (LSTM) unit [29]. The LSTM unit, with N_C hidden cells, captures the temporal dependencies in the CSI by maintaining a hidden state $\mathbf{h}_v \in \mathbb{R}^{N_C}$ and a cell state $\mathbf{c}_v \in \mathbb{R}^{N_C}$. In frame- x , the LSTM unit updates these states based on the current CSI feature vector $\mathbf{z}_{\text{in},v}^{[x]}$ and the previous states $(\mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]})$:

$$(\mathbf{h}_v^{[x]}, \mathbf{c}_v^{[x]}) = f_{\text{LSTM}} \left(\mathbf{z}_{\text{in},v}^{[x]}, \mathbf{h}_v^{[x-1]}, \mathbf{c}_v^{[x-1]}; \boldsymbol{\theta}_{\text{LSTM}} \right), \quad (29)$$

where f_{LSTM} denotes the LSTM transformations and $\boldsymbol{\theta}_{\text{LSTM}}$ denotes the corresponding weights.

3) *Per-BS Output Heads*: The backbone hidden state $\mathbf{h}_v^{[x]}$ is fed into M micro-BS-specific output heads, one for each micro BS:

$$\text{Pred}_{m,v}^{[x+1]} = f_{\text{hd}} \left(\mathbf{h}_v^{[x]}; \boldsymbol{\theta}_{\text{hd},m} \right), \quad \forall m \in \{1, 2, \dots, M\}, \quad (30)$$

where f_{hd} represents the common head structure, $\boldsymbol{\theta}_{\text{hd},m}$ denotes the weights of the m -th head, and $\text{Pred}_{m,v}^{[x+1]}$ is the prediction for the link between micro BS- m and vehicle- v in frame- $(x+1)$. Each head comprises N_H stacked vector ResBlock units [30] with feature dimension N_C , followed by a task-specific final layer. For beam-pair prediction, the final layer has dimension $M_T M_R$, whereas for channel-gain prediction it has dimension 1.

4) *Output Prediction*: Although the beam-pair and channel-gain prediction share the same NN architecture, they are implemented as distinct models with independently trained weights:

1. **Beam Prediction Model**: The optimal beam-pair prediction is formulated as a multi-class classification task. The final layer of each output head employs a softmax activation function to generate a predicted probability distribution $\tilde{\mathcal{P}}_{m,v}^{[x+1]}$ over the $M_T M_R$ beam-pairs. The model is trained with a cross-entropy loss, using a one-hot label whose non-zero entry indicates the index of the optimal beam-pair that maximizes the desired-link gain in frame- $(x+1)$.
2. **Gain Prediction Models**: The channel-gain prediction is regarded as a regression task. For each micro BS-vehicle pair- (m, v) , we train two separate NN models with identical architecture but independent weights: one for the optimal desired-link gain and one for the interference-link gain. In the desired-link model, each head outputs a scalar $\tilde{G}_{m,v}^{[x+1]}$ to approximate the frame-averaged optimal desired-link gain

$$G_{\text{opt},m,v}^{[x+1]} \triangleq \max_{\mathbf{w}_T^{[x+1]}, \mathbf{w}_R^{[x+1]}} G_{m,v}^{[x+1]}, \quad (31)$$

where $G_{m,v}^{[x+1]}$ is obtained by averaging the instantaneous desired-link gains in Eq. (9) over all slots in frame $(x+1)$. The interference-link model is similarly defined: its final layer outputs a scalar $\tilde{G}_{1,m,v}^{[x+1]}$ with label $G_{1,m,v}^{[x+1]}$, given by the frame-average of the instantaneous interference-link gains in Eq. (10). Both models are trained using a mean square error (MSE) loss function, with predictions and labels represented in normalized dB.

The predictions $\{\tilde{\mathcal{P}}_{m,v}^{[x+1]}, \tilde{G}_{m,v}^{[x+1]}, \tilde{G}_{1,m,v}^{[x+1]}\}_{m=1}^M$ reported by each vehicle- v to the macro BS serve as key inputs to the subsequent PHO and BF stages. For the macro BS ($m = 0$), whose low-frequency link does not involve beam alignment or inter-cell interference, only the frame-averaged desired-link gain $\tilde{G}_{0,v}^{[x+1]}$ is required. Owing to the slow variation of the macro-channel, this gain is approximated by the channel gain estimated from control-plane measurements in the current frame, without NN-based prediction.

B. Generalized Assignment-based Proactive Handover

This subsection proposes a generalized assignment-based proactive handover (GAP-HO) algorithm, detailed in Alg. 1, which determines the data-plane connection state for the upcoming frame. Its primary objective is to minimize the per-frame HetNet transmit power while satisfying both the latency-constrained QoS requirements of vehicles and the RB-capacity constraints of BSs.

To ensure queue stability, the expected frame-averaged receive rate $\mathbb{E}[\bar{r}_v^{[x]}]$ should be no less than the data arrival rate λ_v . The following theorem establishes a lower bound on the frame-averaged number of RBs required to satisfy this condition in the interference-free case.

Theorem 1. Consider vehicle- v associated with BS- m in the x -th frame. Given $G_{m,v}^{[x]}$, λ_v , B_m , p_m , F_m and N_0 , the frame-averaged number of allocated RBs, denoted by $\bar{k}_{m,v}^{[x]}$, required to ensure $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$ satisfies

$$\bar{k}_{m,v}^{[x]} \geq \lambda_v \left[B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \right]^{-1}, \quad (32)$$

where $\bar{k}_{m,v}^{[x]} \triangleq N_{\text{HO}}^{-1} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i)$.

Proof. Since vehicle- v is associated with BS- m in frame- x , we have $u_{m,v}^{[x]} = 1$ and $u_{m',v}^{[x]} = 0$ for all $m' \neq m$. Combining (11), (18), (19), and (21), the frame-averaged receive rate is

$$\bar{r}_v^{[x]} = \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right). \quad (33)$$

For any constant $C > 0$, the function $f(z) = \log_2(1 + Cz)$ is concave on $z > 0$. By Jensen's inequality,

$$\mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \leq \log_2 \left(1 + \frac{p_m \mathbb{E}[g_{m,v}^{[x]}(i)]}{F_m N_0 B_m} \right). \quad (34)$$

Moreover, by (9), $\mathbb{E}[g_{m,v}^{[x]}(i)] = G_{m,v}^{[x]}$. Taking expectation of $\bar{r}_v^{[x]}$ and using (34) yield

$$\begin{aligned}\mathbb{E}[\bar{r}_v^{[x]}] &= \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \mathbb{E} \left[\log_2 \left(1 + \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m} \right) \right] \\ &\leq \frac{1}{N_{\text{HO}}} \sum_{i \in \mathcal{N}_{\text{HO}}} k_{m,v}^{[x]}(i) B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right) \\ &= \bar{k}_{m,v}^{[x]} B_m \log_2 \left(1 + \frac{p_m G_{m,v}^{[x]}}{F_m N_0 B_m} \right).\end{aligned}\quad (35)$$

Combining with $\mathbb{E}[\bar{r}_v^{[x]}] \geq \lambda_v$, we obtain the lower bound in (32). This completes the proof. $\square$

Based on Theorem 1, we now construct frame-averaged RB-demand estimates for each BS-vehicle pair- (m, v) in frame- $(x+1)$. For the macro BS ($m = 0$), whose low-frequency link is free of inter-cell interference and beam-alignment overhead, the required frame-averaged number of RBs for vehicle- v is estimated as

$$\tilde{k}_{0,v}^{[x+1]} = \lambda_v \left[B_0 \log_2 \left(1 + \frac{p_0 \tilde{G}_{0,v}^{[x+1]}}{F_0 N_0 B_0} \right) \right]^{-1}, \quad (36)$$

where $\tilde{G}_{0,v}^{[x+1]}$ is estimated from control-plane measurements in the current frame.

For each micro BS ($m > 0$), the lower bound in Theorem 1 is extended to account for inter-cell interference and beam-alignment overhead, while enhancing robustness against occasional large NN prediction errors. Given the NN-predicted frame-averaged gains for frame- $(x+1)$, each gain is smoothed by weighted averaging with the corresponding gain measured in frame- x (when available); if no measurement exists, the NN prediction is used directly. With a slight abuse of notation, we still denote these robustness-enhanced gain estimates by $\tilde{G}_{m,v}^{[x+1]}$ and $\tilde{G}_{1,m',v}^{[x+1]}$. The effective SINR for vehicle- v served by BS- m in frame- $(x+1)$ is then approximated as

$$\widetilde{\text{SINR}}_{m,v}^{[x+1]} = \frac{p_m \tilde{G}_{m,v}^{[x+1]}}{F_m N_0 B_m + \sum_{m' \neq 0, m} \frac{\tilde{k}_{m',v}^{[x+1]}}{K_\mu} p_{m'} \tilde{G}_{1,m',v}^{[x+1]}}, \quad (37)$$

where $\tilde{k}_{m',v}^{[x+1]}$ denotes the estimated frame-averaged RB usage of BS- m' in frame- $(x+1)$ and K_μ is the RB capacity of each micro BS. To capture the beam-training overhead, we conservatively assume that at most M_P beam-search attempts are performed per slot, so that the effective data-transmission fraction per slot is $(1 - \eta M_P)$. Therefore, a conservative estimate of the required frame-averaged number of RBs is

$$\tilde{k}_{m,v}^{[x+1]} = \frac{\lambda_v}{(1 - \eta M_P) B_m \log_2 (1 + \widetilde{\text{SINR}}_{m,v}^{[x+1]})}, \quad m > 0. \quad (38)$$

Since the interference term in (37) depends on the unknown per-BS RB usages $\{\tilde{k}_m^{[x+1]}\}$, we jointly compute the RB demands $\{\tilde{k}_{m,v}^{[x+1]}\}$ and the BS-vehicle associations $\{u_{m,v}^{[x+1]}\}$ via

Algorithm 1 Generalized Assignment-based Proactive Handover (GAP-HO)

Input: $\tilde{G}_{m,v}^{[x+1]}, \tilde{G}_{1,m',v}^{[x+1]}, \lambda_v, p_m, B_m, F_m, K_m, N_0, K_\mu, N_{\text{iter}}, M_P, \eta, \zeta, \epsilon;$
Output: $u_{m,v}^{[x+1]};$

- 1: **Initialize** $\tilde{k}_m^{[x+1]} \leftarrow K_\mu$ for all $m > 0$;
- 2: Compute $\tilde{k}_{0,v}^{[x+1]}$ for all $v \in \mathcal{V}^{[x]}$ by (36);
- 3: **for** $n_{\text{iter}} = 1$ **to** N_{iter} **do**
- 4: Compute $\tilde{k}_{m,v}^{[x+1]}$ for each $m > 0$ and $v \in \mathcal{V}^{[x]}$ by (38);
- 5: Form **P2** with $\{\tilde{k}_{m,v}^{[x+1]}\}$;
- 6: **while** continuous relaxation of **P2** is infeasible **do**
- 7: Multiply the right-hand side of (40) by ζ ;
- 8: **end while**
- 9: Apply the GAP approximation algorithm in [31] to solve **P2** and obtain a PHO schedule $\hat{u}_{m,v}^{[x+1]}$;
- 10: Compute the implied per-BS RB usage $\hat{k}_m^{[x+1]} = \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]}$ for all m ;
- 11: **if** $\max_m |\hat{k}_m^{[x+1]} - \tilde{k}_m^{[x+1]}| \leq \epsilon$ **then**
- 12: $\tilde{k}_m^{[x+1]} \leftarrow \hat{k}_m^{[x+1]}$ for all m ;
- 13: **break**;
- 14: **else**
- 15: $\tilde{k}_m^{[x+1]} \leftarrow \hat{k}_m^{[x+1]}$ for all m ;
- 16: **end if**
- 17: **end for**
- 18: **while** $\hat{u}_{m,v}^{[x+1]}$ is not feasible **and** there exists a feasible reassignment **do**
- 19: Execute the greedy feasibility-repair reassignment that yields the minimum power increase;
- 20: **end while**
- 21: $u_{m,v}^{[x+1]} \leftarrow \hat{u}_{m,v}^{[x+1]}$;
- 22: **return**

a fixed-point iteration, as summarized in Alg. 1. Specifically, we initialize $\tilde{k}_m^{[x+1]} = K_\mu$ for all $m > 0$, evaluate $\tilde{k}_{m,v}^{[x+1]}$ using (37) and (38), and then determine $u_{m,v}^{[x+1]}$ by solving the generalized assignment problem (GAP):

$$\mathbf{P2}: \quad \min_{u_{m,v}^{[x+1]}} \sum_{m=0}^M \sum_{v \in \mathcal{V}^{[x]}} p_m \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \quad (39)$$

$$\text{s.t.} \quad \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} u_{m,v}^{[x+1]} \leq K_m, \quad (40)$$

$$\sum_{m=0}^M u_{m,v}^{[x+1]} = 1, \quad (41)$$

$$u_{m,v}^{[x+1]} \in \{0, 1\}. \quad (42)$$

Since **P2** is NP-hard, we apply the polynomial-time approximation algorithm in [31], which applies a min-cost-matching-based rounding to the continuous relaxation solution of GAP, to obtain a near-optimal solution $\hat{u}_{m,v}^{[x+1]}$; if the continuous relaxation of **P2** is infeasible, we progressively relax (40) via the ζ -scaling procedure in Alg. 1 until feasibility is reached. Denoting $V \triangleq |\mathcal{V}^{[x]}|$, the computational complexity of this

approximation algorithm is $O((M+V)V \log V)$. Moreover, as proved in [31], if **P2** is feasible and the optimal objective value is P^* , $\hat{u}_{m,v}^{[x+1]}$ satisfies an objective value no larger than P^* , while the RB-capacity constraints may be violated by at most a factor of two, i.e., $\sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]} \leq 2K_m$. We then update $\tilde{k}_m^{[x+1]} = \sum_{v \in \mathcal{V}^{[x]}} \tilde{k}_{m,v}^{[x+1]} \hat{u}_{m,v}^{[x+1]}$ and iterate until $\{\tilde{k}_m^{[x+1]}\}$ converges or the maximum iteration number N_{iter} (set to 3 in this paper) is reached. Finally, to strictly satisfy (40), a greedy feasibility-repair reassignment procedure offloads some vehicles from overloaded BSs (if exist) to underloaded BSs with minimum power increase; in the worst case, this reassignment incurs $O(MV^2)$ complexity. Therefore, the computational complexity of GAP-HO is $O(N_{\text{iter}}(M+V)V \log V + MV^2)$.

C. Predictive Early-Termination Beamforming

The predictive early-termination beamforming (PET-BF) algorithm in Alg. 2 is executed at the beginning of each time slot to select beam-pairs for the MIMO links between each micro BS and its associated vehicles. PET-BF exploits the NN-based predictions $\tilde{\mathcal{P}}_{m,v}^{[x]}$ and $\tilde{G}_{m,v}^{[x]}$, obtained in the previous frame as described in Section IV-A, to guide a fast beam-training procedure in each slot of the current frame.

For each micro BS- m and its associated vehicle- v , PET-BF first selects the top- M_P beam-pairs with the highest probabilities from the predicted distribution $\tilde{\mathcal{P}}_{m,v}^{[x+1]}$ over $M_T M_R$ candidates. This selection can be implemented via partial selection in $O(M_T M_R \log M_P)$ time. PET-BF then probes these candidates sequentially in descending order and applies an early-termination rule. Once the measured channel gain g_{meas} of a tested beam-pair exceeds the threshold $\tilde{G}_{m,v}^{[x]} - G_{\text{th}}$ in dB, that beam-pair is immediately selected and the search stops. If none of the M_P candidates satisfies the condition, the beam-pair with the largest measured gain among the probed candidates is selected. As a result, the probing-and-comparison cost is upper-bounded by $O(M_P)$ per link per slot.

This predict-then-refine strategy reduces the beam-training overhead $\eta_{m,v}^{[x]}(i)$ while tracking the instantaneous channel variations $g_{m,v}^{[x]}(i)$, thereby enabling robust and low-latency MIMO links for the considered V2X applications.

D. Opportunistic Tiered-Rounding Resource Allocation

For slot-level RA, we adopt an opportunistic tiered-rounding resource allocation (OTR-RA) algorithm, summarized in Alg. 3. In each slot, after PET-BF, each BS- m executes OTR-RA to allocate its RBs among the associated vehicles under the current channel conditions. The key idea is to exploit instantaneous RB efficiency for opportunistic scheduling, while using a queue-aware tiered-rounding rule for RB allocation.

Since the exact inter-cell interference in slot- i is unknown, BS- m approximates the SINR by leveraging the estimated frame-averaged RB usages $\{\tilde{k}_{m'}^{[x]}\}$ obtained from GAP-HO:

$$\widetilde{\text{SINR}}_{m,v}^{[x]}(i) = \frac{p_m g_{m,v}^{[x]}(i)}{F_m N_0 B_m + \sum_{m' \neq 0, m} \frac{\tilde{k}_{m'}^{[x]}}{K_\mu} p_{m'} g_{1,m',v}^{[x]}(i)}. \quad (43)$$

Algorithm 2 Predictive Early-Termination Beamforming (PET-BF)

Input: $\tilde{\mathcal{P}}_{m,v}^{[x]}, \tilde{G}_{m,v}^{[x]}, M_P, G_{\text{th}}, \eta;$
Output: $w_{T,m,v}^{[x]}(i), w_{R,m,v}^{[x]}(i), g_{m,v}^{[x]}(i), \eta_{m,v}^{[x]}(i)$

- 1: Construct candidate beam-pair index set $\mathcal{W}$ by selecting the M_P largest entries of $\tilde{\mathcal{P}}_{m,v}^{[x]}$;
- 2: **Initialize** $w_{\text{best}} \leftarrow \emptyset, g_{m,v}^{[x]}(i) \leftarrow -\infty$ and $\eta_{m,v}^{[x]}(i) \leftarrow 0$;
- 3: **for all** $w \in \mathcal{W}$ (ordered by probability) **do**
- 4: Probe beam-pair w to measure the instantaneous channel gain in dB: $g_{\text{meas}} \leftarrow \text{MeasureBeamGain}_{\text{dB}}(m, v, w)$;
- 5: $\eta_{m,v}^{[x]}(i) \leftarrow \eta_{m,v}^{[x]}(i) + \eta$;
- 6: **if** $g_{\text{meas}} > g_{m,v}^{[x]}(i)$ **then**
- 7: $g_{m,v}^{[x]}(i) \leftarrow g_{\text{meas}}; w_{\text{best}} \leftarrow w$;
- 8: **end if**
- 9: **if** $g_{\text{meas}} > \tilde{G}_{m,v}^{[x]} - G_{\text{th}}$ **then**
- 10: **break**;
- 11: **end if**
- 12: **end for**
- 13: Map w_{best} to the corresponding $w_{T,m,v}^{[x]}(i)$ and $w_{R,m,v}^{[x]}(i)$;
- 14: **return**

Algorithm 3 Opportunistic Tiered-Rounding Resource Allocation (OTR-RA)

Input: $u_{m,v}^{[x]}, g_{m,v}^{[x]}(i), g_{1,m',v}^{[x]}(i), \eta_{m,v}^{[x]}(i), \tilde{k}_{m'}^{[x]}, q_v^{[x]}(i), K_m, B_m, \Delta t, p_m, F_m, N_0, \lambda_v, \tau_{\text{th}};$
Output: $k_{m,v}^{[x]}(i);$

- 1: **Initialize** $\bar{K}_m \leftarrow K_m$ and $\mathcal{V}_m^{[x]} \leftarrow \{v \in \mathcal{V}^{[x]} \mid u_{m,v}^{[x]} = 1\}$;
- 2: Compute $\Gamma_{m,v}^{[x]}(i)$ for all $v \in \mathcal{V}_m^{[x]}$ by (43) and (44);
- 3: Sort $\mathcal{V}_m^{[x]}$ in descending order of $\Gamma_{m,v}^{[x]}(i)$ to obtain $\mathcal{S}_m^{[x]}$;
- 4: **for all** $v \in \mathcal{S}_m^{[x]}$ **do**
- 5: Compute $k_{m,v}^{[x]}(i)$ by (45) and (46);
- 6: $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$;
- 7: **end for**
- 8: **return**

The corresponding RB efficiency is

$$\Gamma_{m,v}^{[x]}(i) \triangleq B_m \log_2(1 + \widetilde{\text{SINR}}_{m,v}^{[x]}(i)). \quad (44)$$

After computing $\Gamma_{m,v}^{[x]}(i)$ for all associated vehicles, BS- m sorts them in descending order of $\Gamma_{m,v}^{[x]}(i)$ so that links with higher instantaneous RB efficiency are served first. The SINR and RB-efficiency evaluation requires $O(MV_m^{[x]})$ operations due to the summation over interfering micro BSs for each associated vehicle, and the sorting costs $O(V_m^{[x]} \log V_m^{[x]})$.

Given this ordered list, OTR-RA allocates RBs sequentially. For each vehicle- v , an ideal continuous RB demand to clear its backlog in slot- i is

$$\bar{k}_{m,v}^{[x]}(i) = \frac{q_v^{[x]}(i)}{(1 - \eta_{m,v}^{[x]}(i)) \Delta t \Gamma_{m,v}^{[x]}(i)}, \quad (45)$$

which is then mapped to an integer RA decision $k_{m,v}^{[x]}(i)$ via a

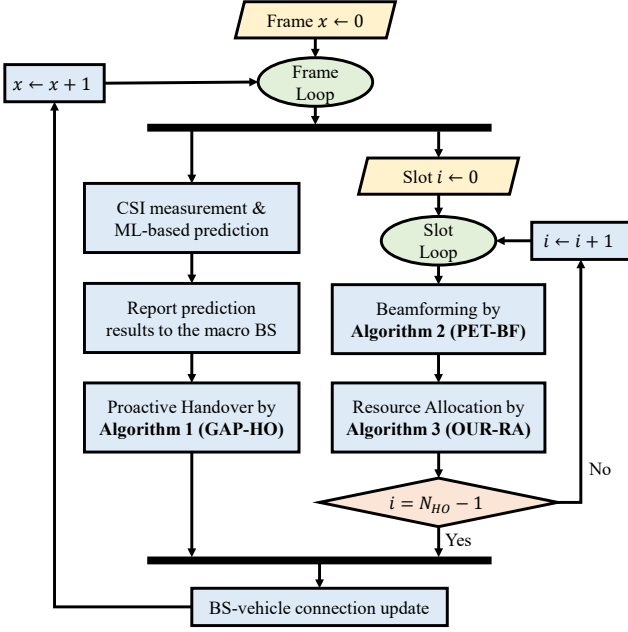


Fig. 2. Flowchart of the MEET-COBRA framework.

queue-aware tiered-rounding rule, subject to the remaining RB budget $\bar{K}_m$:

$$k_{m,v}^{[x]}(i) = \begin{cases} \min\{\lceil \bar{k}_{m,v}^{[x]}(i) \rceil, \bar{K}_m\}, & \text{if } q_v^{[x]}(i) > \frac{\lambda_v \tau_{th}}{2}, \\ \min\{\lfloor \bar{k}_{m,v}^{[x]}(i) \rfloor, \bar{K}_m\}, & \text{otherwise.} \end{cases} \quad (46)$$

Then the budget updates as $\bar{K}_m \leftarrow \bar{K}_m - k_{m,v}^{[x]}(i)$. This tiered-rounding strategy avoids systematic RB waste that would occur if all $\bar{k}_{m,v}^{[x]}(i)$ were rounded up, while also preventing high-urgency users with $\bar{k}_{m,v}^{[x]}(i) < 1$ from repeatedly receiving no RBs under a purely rounding-down rule. The process continues until all the associated vehicles have been scheduled or the RB budget is exhausted. Since each vehicle is processed at most once, the sequential allocation and budget updates cost $O(V_m^{[x]})$. Therefore, the overall per-slot computational complexity of OTR-RA at BS- m is $O(MV_m^{[x]} + V_m^{[x]} \log V_m^{[x]})$, making it a low-complexity RA scheme that balances RB efficiency and queue urgency under per-BS RB-capacity constraints.

E. MEET-COBRA

The proposed MEET-COBRA framework integrates the above modules into a unified two-timescale design, as illustrated in Fig. 2. Frame-level PHO decisions are computed centrally at the macro BS, while slot-level BF and RA are executed locally at each BS.

At the beginning of frame- x , each vehicle performs local NN inference based on its historical CSI measurements to generate next-frame predictions $\{\hat{P}_{m,v}^{[x+1]}, \hat{G}_{m,v}^{[x+1]}, \hat{G}_{1,m,v}^{[x+1]}\}$, which are reported to the macro BS through the control-plane uplink. Using

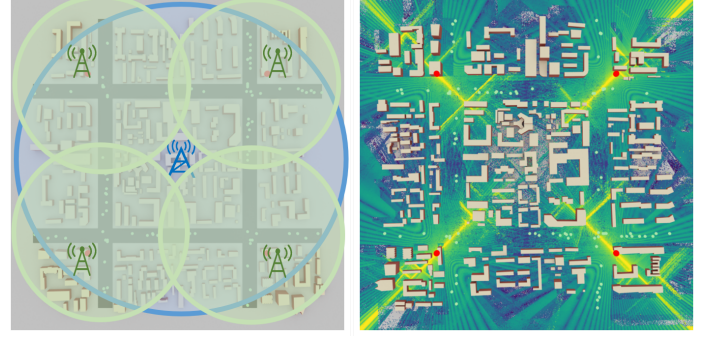


Fig. 3. Simulation scenario of the considered two-tier HetNet. Left: road layout, buildings, and locations/coverage regions of the macro BS (blue) and four micro BSs (green). Right: example Sionna RT ray-tracing-based received-power map at 28 GHz with vertically polarized antennas and micro-BS ULAs oriented toward the area center.

these predictions, the macro BS executes GAP-HO to compute the PHO decisions $u_{m,v}^{[x+1]}$ for the next frame.

Within each slot- i of frame- x , each micro BS- m first applies PET-BF to probe the predicted top- M_P beam-pairs with early termination, thereby obtaining BF vectors $\{\mathbf{w}_{T,m,v}^{[x]}(i), \mathbf{w}_{R,m,v}^{[x]}(i)\}$ and instantaneous link gains $g_{m,v}^{[x]}(i)$ with limited beam-training overhead $\eta_{m,v}^{[x]}(i)$. After that, each BS applies OTR-RA to prioritize users by their RB efficiency $\Gamma_{m,v}^{[x]}(i)$ and perform queue-aware tiered-rounding to obtain integer RB allocations $k_{m,v}^{[x]}(i)$ under the RB-capacity constraints.

Remarkably, ML-based prediction serves as the coupling mechanism that aligns the modeling assumptions across HO, BF, and RA. The gain predictions enable GAP-HO to estimate the next-frame RB demands of each BS-vehicle pair and to solve **P2** based on a forward-looking characterization of the network state, while the beam-pair predictions guide PET-BF to realize high-quality beam alignment under reduced beam-training overhead. This improves the consistency between the link gains assumed in the RB-demand estimation of GAP-HO and the link gains realized during data transmission. Based on these realized link gains, OTR-RA further exploits opportunistic scheduling and queue-aware tiered rounding to enhance latency performance and reduce RB wastage, thereby narrowing the gap between the RB demands predicted by GAP-HO and the realized long-term RB usage. Consequently, MEET-COBRA provides a low-complexity online solution that effectively addresses the original problem **P1**.

V. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed MEET-COBRA framework using a simulation platform built on the SUMO [24] and the Sionna RT [25] simulators. We first describe the simulation setup and parameter configuration, and then present numerical results that compare MEET-COBRA with several baseline schemes in terms of power consumption and latency performance.

TABLE I
SIMULATION PARAMETERS

Global simulation parameters		Macro- and micro-BS parameters		
Parameter	Value	Parameter	Macro BS	Micro BS
Slot duration Δt	1 ms	Carrier frequency f_m	2.8 GHz	28 GHz
PHO frame length N_{HO}	100 slots	Number of RBs K_m	133	66
Pilots per frame N_P	8	RB bandwidth B_m	0.36 MHz	1.44 MHz
Rician factor K_R	3	Per-RB transmit power p_m	1 W	0.2 W
Beam-search overhead η	2/112	Noise figure F_m	5 dB	10 dB
Latency threshold τ_{th}	20 ms	Channel simulation	3GPP TR 38.901 UMa	Ray-tracing by Sionna RT [25]
Noise power spectral density N_0	-174 dBm/Hz	Antenna number (Tx&Rx)	1&1	32&8

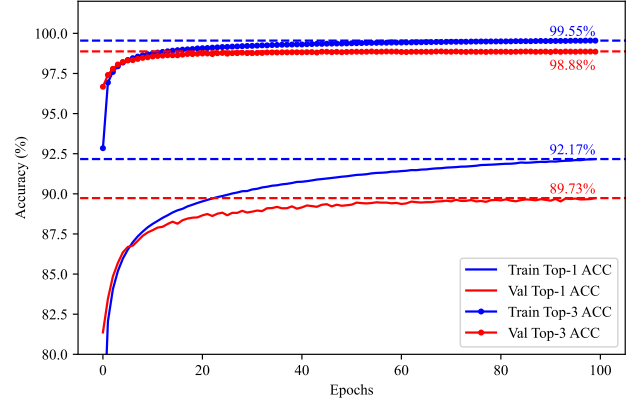
A. Simulation Setup

We consider a $1000 \text{ m} \times 1000 \text{ m}$ urban HetNet scenario shown in Fig. 3, where one macro BS is deployed at the center (0,0) and four micro BSs are located at $(\pm 300, \pm 300) \text{ m}$. The road topology consists of four intersections centered at $(\pm 250, \pm 250) \text{ m}$, and each road segment is bidirectional with three lanes per direction. Each micro BS employs a ULA oriented toward the scenario center. All antennas are vertically polarized. Realistic vehicular mobility trajectories are first generated using the SUMO traffic simulator; based on these trajectories, Sionna RT simulator is used to obtain the time-varying MIMO channel matrices between each vehicle and each micro BS, while the large-scale channel gains between vehicles and the macro BS are generated according to the 3GPP TR 38.901 urban macro model [32]. Unless otherwise stated, the main simulation parameters are summarized in Table I.

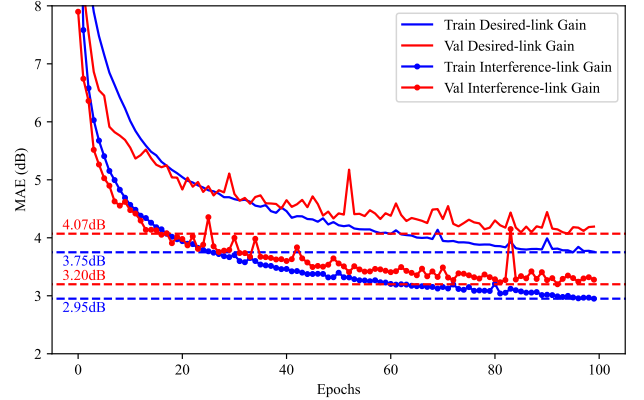
B. Performance of ML-Based Prediction

We evaluate the NN models in Section IV-A for beam-pair and gain prediction. We set $N_C = 128$ for the LSTM unit and the vector ResBlock, with $N_H = 4$ vector ResBlocks per output head. The dataset is built from 6000 frames of Sionna RT-generated channel traces, with about 136 active vehicles per frame on average. To avoid information leakage, the data samples used for training and validation are taken from non-overlapping time intervals. Each sample consists of the historical CSI of a vehicle over 1-10 consecutive frames, together with labels of the optimal beam-pair indices and the corresponding desired-link and interference-link gains for the next frame. The samples are randomly split into training (70%) and validation (30%) sets. All models are trained offline for 100 epochs using the AdamW [33] optimizer with a learning rate of 10^{-3} and a weight decay of 10^{-4} .

Fig. 4 illustrates the training and validation performance of the beam and gain prediction models. For the beam prediction model, the top-1 accuracy on the validation set converges to about 89.7%, while the top-3 accuracy reaches approximately 98.9%, with a small gap between the training and validation curves, indicating good generalization. Besides, we also evaluate the top- M_P reliability of the beam prediction, which exceeds 99% for $M_P \geq 4$ and 99.9% for $M_P \geq 18$, implying that the optimal beam-pair is almost always contained in a short candidate list. For the gain prediction models, the validation mean absolute error (MAE) stabilizes around 4.1 dB for the



(a) Beam prediction model.



(b) Gain prediction models.

Fig. 4. Training and validation performance of the proposed NN models: (a) beam prediction model (top-1/top-3 accuracy); (b) optimal desired-link and interference-link gain prediction models (MAE).

optimal desired-link gain and 3.2 dB for the interference-link gain, and the training/validation curves nearly overlap. These results confirm that the NN models provide sufficiently accurate beam and gain predictions to support MEET-COBRA.

C. Benchmark Schemes

To assess the performance of MEET-COBRA, we compare it with the following benchmark schemes:

1) Oracle-MC: MEET-COBRA with perfect beam and gain knowledge, where all ML predictions are replaced by

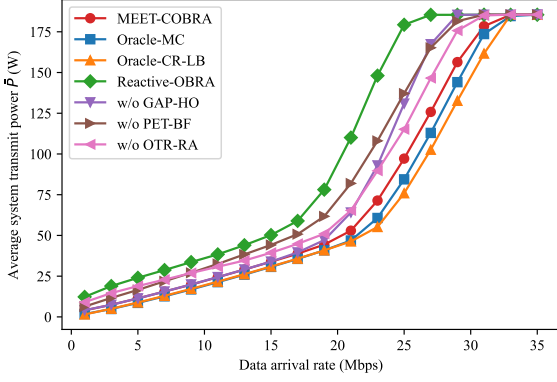


Fig. 5. Average system transmit power $\bar{P}$ versus the data arrival rate λ , where λ is swept over $\{1, 3, \dots, 35\}$ Mbps with $M_P = 5$ and other parameters given in Table I.

their ground-truth values from the simulator. It quantifies the performance loss due to prediction errors.

2) Oracle-CR-LB: A non-implementable power lower bound obtained by solving the continuous relaxation of **P2** (i.e., relaxing $u_{m,v}^{[x+1]} \in \{0, 1\}$ to $[0, 1]$) with perfect beam and gain knowledge. If the relaxed problem is infeasible, the full-load power ceiling is reported. No latency metric is reported for this unrealistic scheme.

3) Reactive-OBRA: A reactive HetNet baseline without ML prediction or proactive coordination. HO is received signal strength (RSS)-based. When the serving BS is overloaded, users are offloaded to the nearest available BS [34]. BF uses a random opportunistic search inspired by [35]: after HO it probes M_P beam-pairs and keeps randomly probing thereafter, updating the serving beam only when a better one is found. RA estimates RB efficiency assuming full-load interfering micro BSs and applies stochastic rounding for integer RB allocation.

4) Ablations: *w/o GAP-HO*, *w/o PET-BF*, and *w/o OTR-RA*, where the corresponding module in MEET-COBRA is replaced by its Reactive-OBRA counterpart, while keeping the other two modules unchanged.

D. Performance Comparison

This subsection compares MEET-COBRA with all benchmark schemes under the same mobility traces and traffic statistics. For each simulation setting, we run the system for a 30 s horizon (i.e., 300 frames) and report metrics time-averaged over this horizon.

Fig. 5 and Fig. 6 compares the average system transmit power $\bar{P}$ and the latency constraint violation probability U under different data arrival rates λ (with $M_P = 5$ and other parameters in Table I). To facilitate interpretation, we partition the network load into three regimes using $\lambda = 20$ and 30 Mbps as coarse thresholds, and summarize the key observations as follows:

- 1) *Light-load regime* ($\lambda < 20$ Mbps): $\bar{P}$ increases approximately linearly with λ and latency violations are

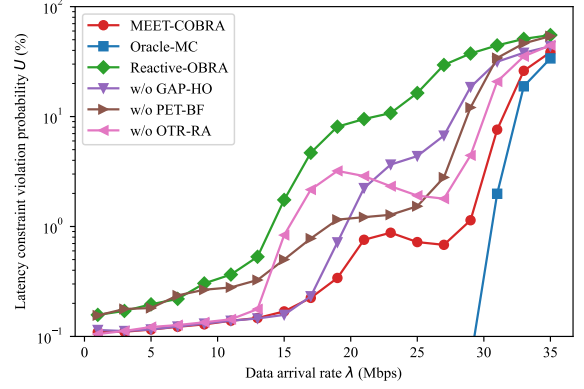


Fig. 6. Latency constraint violation probability U versus the data arrival rate λ , where λ is swept over $\{1, 3, \dots, 35\}$ Mbps with $M_P = 5$ and other parameters given in Table I.

negligible. For instance, at $\lambda = 19$ Mbps, MEET-COBRA achieves $\bar{P} = 44.38$ W with $U = 0.34\%$, approaching Oracle-MC (40.86 W), which indicates only a marginal performance loss due to NN prediction errors. Moreover, Oracle-MC nearly coincides with Oracle-CR-LB in this regime, suggesting that, when perfect CSI/beam knowledge eliminates the prediction errors, the resulting performance is close to the theoretical power lower bound; this shows the near-optimality of the proposed MEET-COBRA framework under light loads.

- 2) *Medium-load regime* ($20 \leq \lambda < 30$ Mbps): RB-capacity constraints become active for some micro BSs and load imbalance emerges. GAP-HO performs global BS-vehicle association optimization, thereby balancing per-BS RB usage and maintaining low $\bar{P}$ and U . For example, at $\lambda = 27$ Mbps, MEET-COBRA achieves $(\bar{P}, U) = (125.82 \text{ W}, 0.68\%)$, whereas Reactive-OBRA already reaches the RB-capacity-imposed power ceiling (about 185.4 W) and suffers a much higher $U = 29.47\%$. Moreover, Fig. 6 exhibits a non-monotonic behavior of U in this regime, where several curves (especially MEET-COBRA, w/o OTR-RA, and w/o PET-BF) temporarily flatten or slightly decrease before rising again. This mainly stems from GAP-HO offloading vehicles from some RB-limited micro BSs to the macro BS, thereby reducing network congestion and inter-cell interference.
- 3) *Heavy-load regime* ($\lambda \geq 30$ Mbps): The network approaches full RB utilization and $\bar{P}$ saturates around the power ceiling, while U increases rapidly as the offered load exceeds the service capability. At $\lambda = 33$ Mbps, MEET-COBRA reaches $\bar{P} = 185.4$ W with $U = 26.08\%$, whereas Reactive-OBRA exhibits a substantially higher violation probability ($U = 50.86\%$).

The ablation results further reveal the distinct roles of GAP-HO, PET-BF, and OTR-RA across different network load states. Under light loads, w/o GAP-HO almost overlaps with

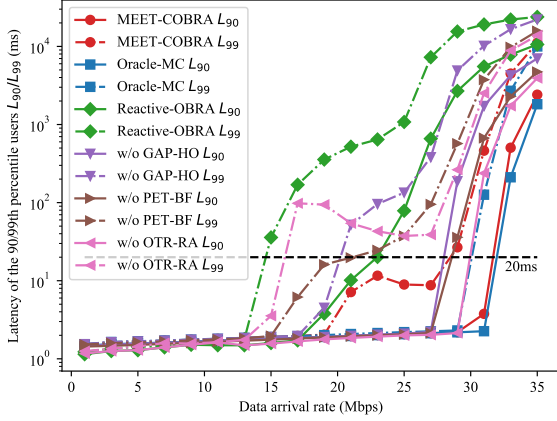


Fig. 7. The 90th- and 99th-percentile user latency versus the data arrival rate λ , where the horizontal dashed line denotes the latency threshold $\tau_{th} = 20$ ms.

MEET-COBRA, suggesting that the near-optimal association is essentially to connect each vehicle to its best micro BS when the RB budgets of micro BSs are sufficient. As the load increases, the performance loss of w/o GAP-HO grows rapidly and eventually becomes the most dominant among all ablation schemes, showing that global association optimization is crucial for load balancing once RB-capacity constraints become active. Moreover, among the three ablation schemes, w/o OTR-RA causes the largest power increase for $\lambda < 10$ Mbps, since conservative interference estimation (assuming full RB utilization for all interfering BSs) and stochastic rounding for RB allocation can lead to substantial resource wastage when the network is lightly loaded, whereas OTR-RA mitigates such wastage via more accurate interference estimation and queue-aware tiered-rounding. Besides, w/o PET-BF degrades network performance across all load regimes, since both BF gain and beam-training overhead continuously impact link efficiency. PET-BF uses NN predictions with early termination to improve the BF gain while reducing training overhead, yielding consistent improvements irrespective of the traffic load.

Fig. 7 reports the 90th/99th-percentile user latency, denoted by L_{90}/L_{99} , and Fig. 8 reports the macro-BS association ratio ρ_0 (in %), i.e., the percentage of vehicles associated with BS-0. Under light loads, all schemes exhibit small L_{90}/L_{99} and almost zero ρ_0 (e.g., MEET-COBRA achieves $\rho_0 = 0.03\%$ at $\lambda = 19$ Mbps), which is consistent with the observation that the RB budgets of each BS are sufficient and most vehicles can be served by favorable micro-BS links.

As the load increases, the tail-latency behavior starts to differentiate and aligns well with the violation probability trends in Fig. 6. A salient metric is the first λ at which L_{99} exceeds the latency threshold $\tau_{th} = 20$ ms: it occurs at $\lambda = 29$ Mbps for MEET-COBRA, whereas it happens much earlier for Reactive-OBRA ($\lambda = 15$ Mbps), w/o OTR-RA ($\lambda = 17$ Mbps), w/o GAP-HO ($\lambda = 21$ Mbps), and w/o PET-BF ($\lambda = 23$ Mbps). This confirms that MEET-COBRA

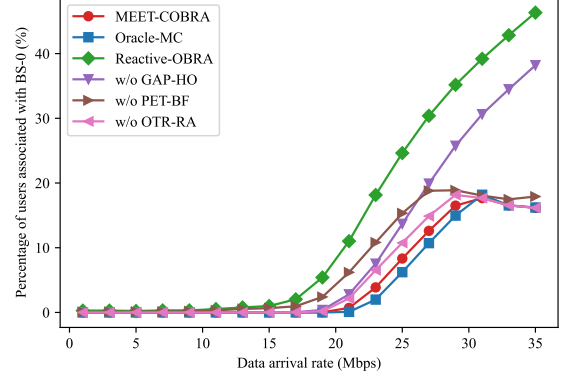


Fig. 8. Percentage of vehicles associated with the macro BS (BS-0) versus the data arrival rate λ .

significantly delays the onset of tail latency degradation.

Fig. 8 provides interpretable evidence on how each scheme redistributes load. At $\lambda = 29$ Mbps, MEET-COBRA maintains a moderate macro-BS association ratio $\rho_0 = 16.50\%$ with only a mild tail-latency exceedance ($L_{99} = 26.76$ ms, $U = 1.14\%$). By contrast, Reactive-OBRA exhibits a much higher $\rho_0 = 35.18\%$ with a tail-latency collapse ($L_{99} = 1.55 \times 10^4$ ms, $U = 37.46\%$), while w/o GAP-HO increases ρ_0 to 25.75% and enlarges L_{99} to 4.91×10^3 ms with $U = 18.54\%$. Moreover, the mild non-monotonicity of L_{99} in the medium-load regime in Fig. 7 is similar to that of U in Fig. 6, which coincides with the initial rise of ρ_0 in Fig. 8. These results demonstrate that the globally optimized PHO of GAP-HO effectively balances network load under RB-capacity constraints, thereby delaying congestion build-up and mitigating extreme-tail latency.

Finally, the joint use of Fig. 7 and Fig. 8 helps disentangle the roles of PET-BF and OTR-RA beyond association decisions. At $\lambda = 29$ Mbps, w/o PET-BF and w/o OTR-RA have ρ_0 comparable to MEET-COBRA (18.88% and 18.16% versus 16.50%), yet their tail latency is much worse ($L_{99} = 5.71 \times 10^2$ ms and 2.64×10^2 ms, respectively). Notably, w/o PET-BF also enlarges L_{90} to 35.96 ms, indicating a latency performance degradation due to inefficient beam search without prediction-assisted candidate reduction and early termination, whereas w/o OTR-RA keeps L_{90} close to MEET-COBRA (2.14 ms versus 2.19 ms) but significantly enlarges L_{99} , suggesting that queue-aware tiered rounding mainly protects urgent users and mitigates extreme-tail violations.

VI. CONCLUSION

This paper proposed MEET-COBRA, a ML-based energy-efficient proactive coordination framework for jointly optimizing PHO, BF and RA in HetNets. By leveraging ML-based prediction of beam-pair candidates and channel gains, MEET-COBRA provides a low-complexity online approach to efficiently address the two-timescale SINLP for COBRA.

Simulations driven by SUMO mobility traces and Sionna RT channel data validate the effectiveness of MEET-COBRA.

The ML-driven proactive optimization mechanism delivers substantial gains: the predicted beam-pair and channel gain information enables efficient global PHO optimization, reduces beam-search overhead, and mitigates RB wastage and latency constraint violations caused by suboptimal BF and RA decisions, thereby addressing the computational challenges of the two-timescale SINLP in practical deployments. Notably, MEET-COBRA effectively balances system transmit power and latency reliability: at an illustrative medium-to-high load operating point ($\lambda = 27$ Mbps), it reduces the average system transmit power by approximately 32% compared with the Reactive-OBRA baseline, while decreasing the latency constraint violation probability from 29.47% to 0.68%.

Potential future work includes extending MEET-COBRA to multi-macro BS deployments, incorporating more realistic HO latency and signaling overhead, and developing online adaptation mechanisms for the prediction models and algorithm hyper-parameters to enhance robustness under non-stationary traffic and radio environments.

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